Sahaj Bhadja

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Summary

Software engineer with 2+ years building **Python/TypeScript full-stack systems**, **FastAPI-style backend services**, and data-intensive workflows, with strong delivery across cloud deployment, ETL reliability, and cross-functional collaboration with product and analytics teams.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Structures, Operating Systems, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built **Python-first platform services** for 500+ users, reducing request latency from **1.2s to 350ms** by optimizing execution paths, tuning SQL access patterns, and improving responsiveness for interactive analytical workflows.
- Implemented **automated validation and ETL checks** for 5,000+ monthly jobs, reducing downstream failures and accelerating incident triage through earlier detection of workflow regressions, data-quality drift, and runtime anomalies.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed **full-stack features and APIs** handling 10,000+ daily requests, translating product requirements into maintainable backend services and responsive interfaces while improving delivery predictability through clearer implementation ownership and sequencing.
- Improved **release velocity** through CI/CD automation and structured defect triage, reducing turnaround by **60%**, lowering manual QA friction, and enabling safer, faster deployments across parallel engineering workstreams.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **large-scale experimentation pipelines** across 100,000+ trajectory records, enabling reproducible algorithm evaluation and high-confidence benchmarking that accelerated iteration while improving signal quality for debugging model-driven decision workflows.
- Produced **technical reports and test scenarios** translating findings into actionable implementation guidance, improving cross-team alignment, reducing handoff ambiguity, and strengthening engineering quality as prototypes matured toward production readiness.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **API and ETL microservices** using AWS Glue and S3, improving response time by **30%** while supporting distributed integrations, resilient data movement, and stable delivery across enterprise production workloads.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and improving confidence in frequent releases across coordinated service components and teams.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Developed **MERN and WebSocket modules** for 2M+ synchronized records across eight workflows, improving throughput consistency, reducing stale-session behavior, and strengthening real-time responsiveness for high-volume user-facing operational processes.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing environment drift that previously slowed validation, troubleshooting, and cross-team handoffs.

Skills

Languages: **Python**, **TypeScript**, JavaScript, SQL, Java, C++, Bash
Frontend & UI: TypeScript, React, component-driven interfaces, responsive product workflows
Backend & Data: Python APIs, REST patterns, ETL pipelines, SQL tuning, distributed integration workflows
Cloud & Platform: AWS, Docker, Kubernetes, Helm basics, Terraform basics, CI/CD, Linux/UNIX
Delivery: Automated testing, production debugging, technical documentation, Agile/Scrum, cross-functional collaboration

Projects

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Built a **containerized REST processing API** on AWS CodePipeline and ECS/Fargate with automated validation gates, improving reliability, reducing deployment variance, and streamlining operational troubleshooting across production ETL workflows.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Developed a **schema-aware query assistant** spanning 10+ datasets with explainable outputs and validation checks, improving analyst speed on ad-hoc tasks while reducing malformed SQL generation across exploratory analysis workflows.

CaptionCraft Code | *Python, FastAPI, React, LLM*
- Implemented an **LLM-assisted captioning workflow** with evaluation hooks and configurable prompts, improving iteration speed for user-facing generation features while strengthening observability of output quality across rapid experimentation cycles.